

EC 551 Advanced Digital Design with Verilog and FPGAs

Fall 2026, Prof. Douglas Densmore

Homework #1

Due: 10/1

1. Do the following HW problems in Chapter 2 of the textbook: 1, 5, 8, 10d, 12, 13
2. Do the following HW problems in Chapter 3 of the textbook: 3, 6, 7, 14, 15
3. Convert the following numbers into decimal. Show all intermediate steps used for conversion (assume unsigned numbers).

1010100_2

1111000_2

3221_4

$AA5F_{16}$

4. Express the following numbers using 2's complement binary representation (8 bits) and add the two numbers. Show all intermediate steps used for performing addition operations.

(a) -47_{10}

(b) -123_4

5. Use the Tablular method to minimize the following expression. Please include the following steps:
 - A. Identify the prime implicants
 - B. Create a table with minterms assigned to the prime implicants
 - C. Identify and select the essential prime implicants
 - D. Select the remaining prime implicants and create a minimal minterm cover

During this process, you should be showing the table transformations that occur as rows and columns are removed.

$$z(a,b,c,d) = a'b'(c'd+c'd') + a'bc'd + ab(c+d)+ac'd$$

6. Please provide 3-4 sentences on your current project ideas. Do you have any team members in mind? Are you going to try to tie this work in with another project or class?